SUDHEER SINGH

New Delhi, India | +91-9826235849 | singhsudheer215@gmail.com
linkedin.com/in/sudheer-singh-975889195 | github.com/sudheer215s

PROFESSIONAL SUMMARY

Backend Developer with 3+ years of experience building and operating high-availability services for large-scale telecom systems handling millions of daily subscriber lookups. Skilled in Python, TypeScript, SQL, FastAPI, NestJS, PostgreSQL, Kafka, Redis, Docker, and Kubernetes. Delivered **40% faster critical queries**, **99.99% database availability**, sub-30-second recovery, and a **70% reduction in deployment effort**.

PROFESSIONAL EXPERIENCE

Scientist-B — Backend Engineer | Centre for Development of Telematics (C-DOT) July 2024 – Present
New Delhi, India

- Develop and operate backend services for the national IMS/ENUM call-routing system, resolving **millions of subscriber lookups daily** with sub-second response times.
- Designed and deployed a highly available PostgreSQL cluster with streaming replication and automatic failover, achieving **99.99% uptime**, sub-30-second recovery, and zero unplanned outages since launch.
- Optimized critical IMS/ENUM queries through query-plan analysis and index redesign, improving performance by **40% under peak load**.
- Implemented database monitoring and alerting that reduced production issue detection from user-reported incidents to minutes.
- Designed and developed a Number Portability Server for telecom routing workflows using the SS7/SIGTRAN protocol stack.

Research Engineer — Backend & Systems | Centre for Development of Telematics (C-DOT) July 2023 – June 2024
New Delhi, India

- Implemented NSSAAF service interfaces for the 5G core, translating requirements from 3GPP TS 23.501 and TS 29.509 into production API contracts; passed compliance validation on the first submission with zero non-conformances.
- Operated Ubuntu Linux servers and PostgreSQL databases supporting live telecom services, maintaining **99.9% availability** throughout the year.
- Automated deployment and configuration workflows with Python and Bash, reducing manual effort by **70%**.
- Implemented authentication controls across multiple 5G network functions, remediating three gaps identified during an internal security audit.

PROJECTS

Distributed Social Media Backend
NestJS, TypeScript, PostgreSQL, Kafka, Redis, gRPC, Elasticsearch, Kubernetes

- Architected a NestJS monorepo with five domain-focused microservices—User, Post, Graph, Notification, and Search—communicating over gRPC.
- Implemented JWT authentication, RBAC, shared guards and interceptors, rate limiting, circuit breakers, and Kafka's transactional outbox pattern for resilient service communication.
- Built Redis-backed timeline fan-out, cross-instance WebSocket notifications, and Elasticsearch search; deployed on Kubernetes with CI/CD, Prometheus/Grafana, and distributed tracing.

ClauseNotebook — Citation-Grounded 3GPP Document Pipeline (In Progress)
Python, CLI, JSON, SHA-256, PyMuPDF4LLM

- Building the ingestion foundation for a citation-grounded 3GPP standards copilot; tracks raw source PDFs in a versioned JSON manifest with metadata and SHA-256 provenance.
- Developed an idempotent Python CLI for repository-contained registration, duplicate protection, atomic manifest updates, and PyMuPDF4LLM conversion to self-describing Markdown for clause-level retrieval.

TranslateLive — Real-Time Voice Translation Backend
Python, FastAPI, WebSocket, Redis, Deepgram, OpenAI, Docker, Kubernetes

- Built an asynchronous voice-processing pipeline for speech recognition, translation, and text-to-speech playback with **under three seconds of end-to-end latency**.
- Used Python asyncio, connection pooling, and Redis session state to support **more than 10 simultaneous WebSocket sessions** without dropped connections and enable horizontal scaling.
- Deployed the service on Kubernetes with continuous integration, integration tests, structured JSON logging, and health checks.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash
Backend: FastAPI, NestJS, Node.js, REST APIs, gRPC, WebSocket, JWT, RBAC
Databases & Messaging: PostgreSQL, Redis, Elasticsearch, Kafka
Infrastructure & Observability: Docker, Kubernetes, Linux, CI/CD, Prometheus, Grafana, distributed tracing
AI-Assisted Development: Claude Code, OpenAI Codex, Grok Build
Core Engineering: Distributed systems, microservices, event-driven architecture, high availability, query optimization, caching, system design

EDUCATION

Bachelor of Technology in Computer Science and Engineering 2019 – 2023
Maulana Azad National Institute of Technology (MANIT), Bhopal